

SECTION A: MULTIPLE CHOICE QUESTIONS - 23 MARKS

READ THE FOLLOWING INSTRUCTIONS

1. YOU HAVE TO START WITH SECTION A, AS YOU HAVE TO HAND IN THE OPTIC READER FORM AFTER 75 MINUTES.
2. As you may not erase wrong answers on the optic reader form, first circle your answers on this paper and only mark the answers on the form when you are certain of your choice.
3. Do all scribbling on the facing page. It will not be marked.
4. Use a soft pencil to write on the optic reader form.
5. Use **SIDE 2** of the optic reader form.
Fill in your personal information in pencil on side 2.
Fill in your student number from top to bottom and then code it.
6. You may start with section B as soon as you have marked your answers on the optic reader form.

AFDELING A : MEERVOUDIGE KEUSEVRAE - 23 PUNTE

LEES DIE VOLGENDE INSTRUKSIES

1. U MOET MET AFDELING A BEGIN WANT U MOET DIE MERKLEESVORM NA 75 MINUTE INHANDIG.
2. U mag nie verkeerde antwoorde uitvee op die merkleesvorm nie. Omkring dus u antwoorde eers in hierdie vraestel en merk die antwoorde op die vorm as u seker is van u keuse.
3. Doen alle krapwerk op die teenblad. Dit word nie nagesien nie.
4. Gebruik 'n sagte potlood om op die merkleesvorm te skryf.
5. Gebruik **KANT 2** van die merkleesvorm.
Vul u persoonlike inligting in potlood in op kant 2.
Vul u studentenommer van bo na onder in en kodeer dit.
6. U kan met afdeling B begin sodra u die antwoorde op die merkleesvorm ingevul het.

Question 1 / Vraag 1

If $f(x) = \sqrt{\frac{1}{x+1}}$, the domain of f is

Indien $f(x) = \sqrt{\frac{1}{x+1}}$, dan is die definisieversameling van f

(a)	$[-1, \infty)$	(b)	$(-\infty, -1) \cup (-1, \infty)$	☒ (c)	$(-1, \infty)$	(d)	$(-\infty, -1) \cup (\frac{1}{2}, \infty)$
(e)	$\mathbb{R}$	(f)	$[-1, \infty]$	(g)	None of these / Geen van hierdie		

Question 2 / Vraag 2

Given / Gegee $f(x) = \begin{cases} 4 - x^2 & \text{if } x \leq 1 \\ mx + b & \text{if } x > 1 \end{cases}$

If the function f is continuous and differentiable on $\mathbb{R}$, then

Indien die funksie f kontinu en differensieerbaar is op $\mathbb{R}$, dan is

(a)	$m = 2; b = 1$	☒ (b)	$m = -2; b = 5$	(c)	$m = -2; b = 1$	(d)	$m = 2; b = 5$
(e)	$m = 2; b = 2$	(f)	$m = 1; b = 1$	(g)	None of these / Geen van hierdie		

Question 3 / Vraag 3

The average rate of change of the function $f(x) = x^2 - x + 1$ on $[0, 3]$ is

Die gemiddelde tempo van verandering van die funksie $f(x) = x^2 - x + 1$ op $[0, 3]$ is

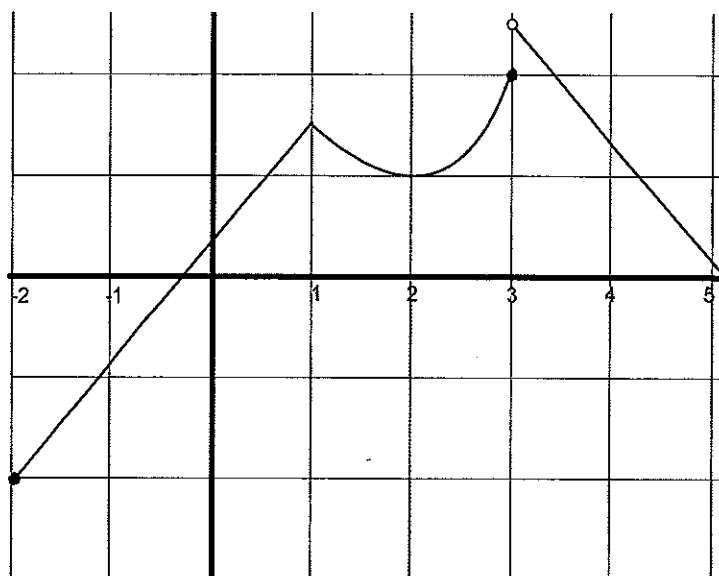
(a)	1	(b)	-1	☒ (c)	2	(d)	-2
(e)	5	(f)	-5	(g)	None of these / Geen van hierdie		

The information below applies to Questions 4 to 7

Die onderstaande inligting is van toepassing op Vrae 4 tot 7

The graph of the function f with domain $[-2, 5]$ is given below.

Die grafiek van die funksie f met definisieversameling $[-2, 5]$ word hieronder gegee.



Question 4 / Vraag 4

At which of the values of $x, x \in \{1, 2, 3\}$, is f continuous? At

By watter waardes van $x, x \in \{1, 2, 3\}$, is f kontinuu? By

(a)	$x = 1, x = 2$ and/en $x = 3$	(b)	$x = 1$ and/en $x = 3$	(c)	only / slegs $x = 1$
(d)	only / slegs $x = 2$	(e)	$x = 2$ and/en $x = 3$		
☒ (f)	$x = 1$ and/en $x = 2$	(g)	None of these / Geen van hierdie		

Question 5 / Vraag 5

At which of the values of $x, x \in \{1, 2, 3\}$, is f differentiable? At

By watter waardes van $x, x \in \{1, 2, 3\}$, is f differensieerbaar? By

(a)	$x = 1, x = 2$ and/en $x = 3$	(b)	$x = 1$ and/en $x = 3$	(c)	only / slegs $x = 1$
☒ (d)	only / slegs $x = 2$	(e)	$x = 2$ and/en $x = 3$		
(f)	$x = 1$ and/en $x = 2$	(g)	None of these / Geen van hierdie		

Question 6 / Vraag 6

Which of the following statement(s) is/are true? / Watter van die volgende bewering(s) is waar?

I. f has an absolute minimum at $x = -2$ / f het 'n absolute minimum by $x = -2$

II. f has an absolute maximum at $x = 3$ / f het 'n absolute maksimum by $x = 3$

III. f has a local minimum at $x = 2$ / f het 'n lokale minimum by $x = 2$

(a)	only / slegs I	(b)	only / slegs II	(c)	only / slegs III
(d)	only / slegs I and/en II	(e)	only / slegs I and/en III		
(f)	only / slegs II and/en III	(g)	None of these / Geen van hierdie		

Question 7 / Vraag 7

Which of the following statement(s) is/are true? / Watter van die volgende bewering(s) is waar?

I. f is continuous from the left at $x = 3$ / f is kontinuu van links in $x = 3$

II. f is continuous from the right at $x = 3$ / f is kontinuu van regs in $x = 3$

III. f is continuous from the left at $x = -2$ / f is kontinuu van links in $x = -2$

(a)	only / slegs I	(b)	only / slegs II	(c)	only / slegs III
(d)	only / slegs I and/en II	(e)	only / slegs I and/en III		
(f)	only / slegs II and/en III	(g)	None of these / Geen van hierdie		

The information below applies to Questions 8 and 9

Die onderstaande inligting is van toepassing op Vrae 8 en 9

Let $S(x) = \sqrt{x}$. Functions f and g are differentiable and the following is known:

Laat $S(x) = \sqrt{x}$. Funksies f en g is differensieerbaar en die volgende is bekend:

$$f(3) = 2, f'(3) = 7, g(3) = 4, g'(3) = 5$$

Question 8 / Vraag 8

$$(fg)'(3) =$$

(a)	35	(b)	30	(c)	28	(d)	10
(e)	38	(f)	27	(g)	None of these / Geen van hierdie		

Question 9 / Vraag 9

$$(S \circ g)'(3) =$$

(a)	1	(b)	$\frac{1}{2}$	(c)	2	(d)	$\frac{5}{4}$
(e)	$\frac{4}{5}$	(f)	3	(g)	None of these / Geen van hierdie		

Question 10 / Vraag 10

If $F(x) = x \sin x$, then $F'(\frac{3\pi}{2}) =$

Indien $F(x) = x \sin x$, dan is $F'(\frac{3\pi}{2}) =$

(a)	0	(b)	1	☒ (c)	-1	(d)	$\frac{3\pi}{2}$
(e)	$-\frac{3\pi}{2}$	(f)	π	(g)	None of these / Geen van hierdie		

Question 11 / Vraag 11

If $y^3x + y^2x^2 = 6x$, then $\frac{dy}{dx} =$

Indien $y^3x + y^2x^2 = 6x$, dan is $\frac{dy}{dx} =$

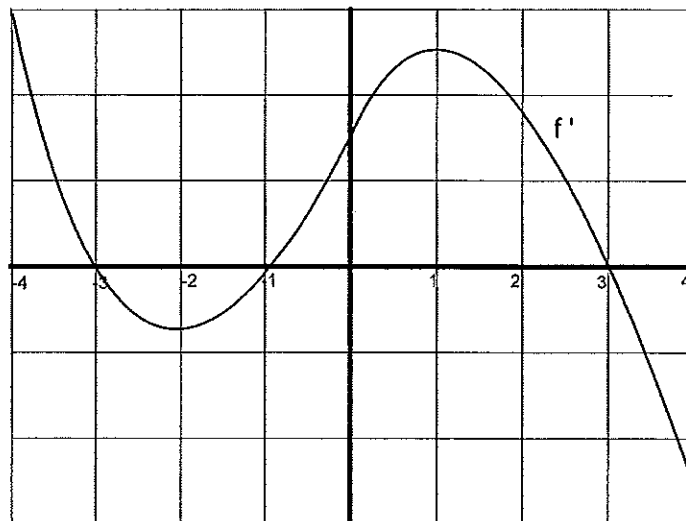
(a)	$\frac{6}{3y^2 + 4xy}$	(b)	$\frac{6 - y^3 - 2xy^2 - 2x^2y}{3xy^2}$	☒ (c)	$\frac{6 - y^3 - 2xy^2}{3xy^2 + 2x^2y}$	(d)	$\frac{6 - 2x^2y}{3xy^2 + 2x^2y}$
(e)	$\frac{6x - 2x^2y}{3xy^2 + 2x^2y}$	(f)	0	(g)	None of these / Geen van hierdie		

The information below applies to Questions 12 and 13

Die onderstaande inligting is van toepassing op Vrae 12 en 13

The graph of the derivative of a function f is given below.

Die grafiek van die afgeleide van 'n funksie f is gegee hieronder.



Question 12 / Vraag 12

Which statement regarding the original function f is true?

Watter stelling met betrekking tot die oorspronklike funksie f is waar?

(a)	f is increasing on the interval $(-2, 1)$ / f is stygend op die interval $(-2, 1)$
(b)	f is increasing on the interval $(3, 4)$ / f is stygend op die interval $(3, 4)$
(c)	f is decreasing on the interval $(1, 4)$ / f is dalend op die interval $(1, 4)$
☒ (d)	f is decreasing on the interval $(-3, -1)$ / f is dalend op die interval $(-3, -1)$
(e)	f has a local maximum at $x = 1$ / f het 'n lokale maksimum by $x = 1$
(f)	None of these / Geen van hierdie

Question 13 / Vraag 13

Which statement regarding the original function f is true?

Watter stelling met betrekking tot die oorspronklike funksie f is waar?

(a)	f has an absolute minimum at $x = -2$ / f het 'n absolute minimum by $x = -2$
(b)	f has a local minimum at $x = -2$ / f het 'n lokale minimum by $x = -2$
☒ (c)	f has an inflection point at $x = -2$ / f het 'n infleksie punt by $x = -2$
(d)	None of these / Geen van hierdie

Question 14 / Vraag 14

Given / Gegee

$$g(x) = x^3 - 3x + 2$$

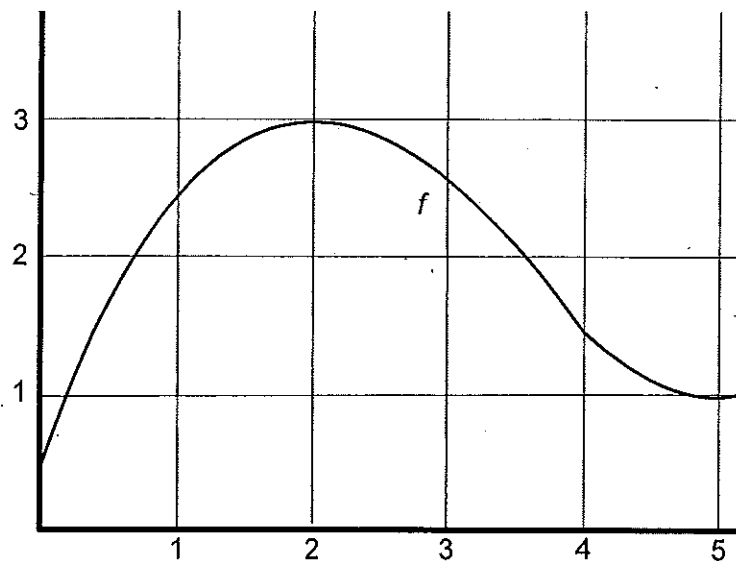
The interval on which the function g is concave downward is given by:

Die interval waarop die funksie g konkaf na onder is word gegee deur:

☒ (a)	$(-\infty, 0)$	(b)	$(-1, \infty)$	(c)	$(1, \infty)$	(d)	$(0, \infty)$
(e)	$(-1, 1)$	(f)	$\mathbb{R}$	(g)	None of these / Geen van hierdie		

Question 15 / Vraag 15

The graph of the function f is given below. Die grafiek van die funksie f word hieronder gegee.



Which statement regarding the function f is true?

Watter stelling met betrekking tot die funksie f is waar?

(a)	$\lim_{h \rightarrow 0} \frac{f(2+h)-f(2)}{h} < f'(4)$
(b)	$\frac{f(5)-f(2)}{5-2} = \frac{2}{3}$
☒ (c)	$f''(1) \leq f''(5)$
(d)	None of these / Geen van hierdie

Question 16 / Vraag 16

Given / Gegee

$$f(x) = (x-2)^4$$

The function f has (an) inflection point(s) at:

Die funksie f het ('n) infleksie punt(e) by:

(a)	2	(b)	0	(c)	$\frac{1}{2}$	(d)	-2
(e)	2 and / en -2	☒ (f)	Does not have points of inflection / Het nie infleksiepunte nie				

Question 17 / Vraag 17

Given that for the function f , continuous on $\mathbb{R}$, the following applies:
Gegee dat vir die funksie f , kontinu op $\mathbb{R}$, die volgende geld:

$$f(7) = f'(7) = 0 \text{ and } f''(7) = 2.$$

Which one of the following statements is true?

Watter een van die volgende bewerings is waar?

☒ (a)	The function has a local minimum at $x = 7$ / Die funksie het 'n lokale minimum by $x = 7$
(b)	The function has a local maximum at $x = 7$ / Die funksie het 'n lokale maksimum by $x = 7$
(c)	The function has an inflection point at $x = 7$ / Die funksie het 'n infleksiepunt by $x = 7$
(d)	None of these / Geen van hierdie

Question 18 / Vraag 18

$$\frac{2 \sinh x}{e^x} =$$

☒ (a)	$1 - e^{-2x}$	(b)	$1 + e^{-x}$	(c)	$1 - e^x$	(d)	$1 + e^x$
(e)	$1 - e^{-x}$	(f)	$1 + e^{-2x}$	(g)	None of these / Geen van hierdie		

Question 19 / Vraag 19

$$\lim_{x \rightarrow \infty} (\sqrt{x} - x) =$$

(a)	∞	(b)	0	(c)	-1	(d)	1
☒ (e)	$-\infty$	(f)	2	(g)	None of these / Geen van hierdie		

Question 20 / Vraag 20

$$\lim_{x \rightarrow \infty} \frac{5x^2}{3x^2 + 10000x} =$$

(a)	0	(b)	$\frac{1}{200}$	(c)	1	☒ (d)	$\frac{5}{3}$
(e)	∞	(f)	$-\infty$	(g)	None of these / Geen van hierdie		

Question 21 / Vraag 21

$$\lim_{x \rightarrow 0^-} \left(\frac{1}{x} + \frac{1}{x^2} \right) =$$

(a)	0	(b)	$\frac{1}{2}$	(c)	1	(d)	2
☒ (e)	∞	(f)	$-\infty$	(g)	None of these / Geen van hierdie		

The information below applies to Questions 22 and 23

Die onderstaande inligting is van toepassing op Vrae 22 en 23.

Let / Laat $F(x) = \frac{3x}{x^2 - 1}$

Question 22 / Vraag 22

The interval(s) on which F increases is

Die interval(le) waarop F styg is

(a)	$(-\infty, -1) \cup (-1, 1) \cup (1, \infty)$	(b)	$\mathbb{R}$	(c)	$(-1, 1)$	(d)	$(-\infty, -1) \cup (1, \infty)$
☒ (e)	$\emptyset$	(f)	$\mathbb{R} \setminus \{1\}$	(g)	None of these / Geen van hierdie		

Question 23 / Vraag 23

Horizontal asymptote(s) of F is

Horisontale asimptot(e) van F is

☒ (a)	$y = 0$	(b)	$y = 1$ and/en $y = -1$	(c)	$y = 2$ and/en $y = -2$	(d)	$y = 2$
(e)	$y = 1$	(f)	No horizontal asymptote exists / Geen horisontale asimptoot bestaan nie				

23

SECTION B: 29 MARKS

READ THE FOLLOWING INSTRUCTIONS

1. Answer all the following questions on this paper in ink.
2. Work done in pencil or in red ink will not be marked.
3. You are not allowed to use correction fluid ("Tipp-Ex").
4. Do all scribbling on the facing page. It will not be marked.
If you need more space for an answer, use the facing page and indicate it clearly by framing it.
5. Show all steps and calculations and explain your answers.

AFDELING B: 29 PUNTE

LEES DIE VOLGENDE INSTRUKSIES

1. Beantwoord die volgende vrae op hierdie vraestel in ink.
2. Werk wat in potlood of rooi ink beantwoord is, word nie nagesien nie.
3. U mag nie korrigeer-vloeistof ("Tipp-Ex") gebruik nie.
4. Doen alle krapwerk op die teenblad. Dit word nie nagesien nie.
As u nog ruimte vir 'n antwoord nodig het, gebruik die teenblad en dui dit duidelik aan deur 'n raam daarom te trek.
5. Toon alle stappe en berekeninge en verduidelik u antwoorde.

Question 1 / Vraag 1

Given / Gegee

$$f(x) = \begin{cases} \frac{x^4-1}{x-1} & \text{if } x < 1 \\ 4 & \text{if } x \geq 1 \end{cases}$$

Determine whether f is continuous at $x = 1$. / Stel vas of f kontinu is by $x = 1$.

$$(i) \quad \lim_{x \rightarrow 1^-} \frac{x^4-1}{x-1} = \lim_{x \rightarrow 1^-} \frac{(x-1)(x+1)(x^2+1)}{x-1} \\ = 4$$

$$(ii) \quad \lim_{x \rightarrow 1^+} 4 = 4$$

$$\Rightarrow \lim_{x \rightarrow 1} f(x) = 4$$

$$(iii) \quad f(1) = 4$$

$$\text{From (i), (ii), (iii)} \quad \lim_{x \rightarrow 1} f(x) = f(1)$$

$\Rightarrow f$ continuous at $x = 1$

Question 2 / Vraag 2

Determine the following derivatives. Do not simplify your answers.

Bepaal die volgende afgeleides. Moenie jou antwoorde vereenvoudig nie.

2.1 $f(x) = \frac{1}{3-x} = (3-x)^{-1}$

$$f'(x) = - (3-x)^{-2} = \frac{+1}{(3-x)^2}$$

1

2.2 $f(t) = \arctan(e^{2t})$ / $f(t) = \text{bgtan}(e^{2t})$

$$f'(t) = \frac{2e^{2t}}{1 + (e^{2t})^2}$$

1

2.3 $f(x) = \frac{3^{\sec x}}{e^{2x}}$ (Use the quotient rule. / Gebruik die kwosiëntreël.)

$$f'(x) = \frac{3^{\sec x} \cdot \ln 3 \cdot \sec x \tan x \cdot e^{2x} - 3^{\sec x} e^{2x}}{(e^{2x})^2}$$

2

2.4 $f(x) = (\ln x)^{\frac{1}{x}}$

let $y = (\ln x)^{\frac{1}{x}}$

$$\Rightarrow \ln y = \frac{1}{x} \ln(\ln x)$$

$$\frac{y'}{y} = -\frac{1}{x^2} \ln(\ln x) + \frac{1}{x} \frac{\frac{1}{x}}{\ln x}$$

$$y' = y \left[-\frac{1}{x^2} \ln(\ln x) + \frac{1}{x^2 \ln x} \right]$$

2

$$2.5 \ y = \ln\left(\frac{(x^2+1)^2}{\arcsin x}\right) / \ y = \ln\left(\frac{(x^2+1)^2}{\lg \sin x}\right)$$

$$\Rightarrow y = 2 \ln(x^2+1) - \ln \arcsin x$$

$$\Rightarrow \frac{dy}{dx} = 2 \frac{2x}{x^2+1} - \frac{\frac{1}{\sqrt{1-x^2}}}{\arcsin x}$$

3

Question 3 / Vraag 3

Assume $\frac{d}{dx} \cos x = -\sin x$. Use this to find $\frac{d}{dx} \sec 3x$. Simplify your answer.

Aanvaar $\frac{d}{dx} \cos x = -\sin x$. Gebruik dit en vind $\frac{d}{dx} \sec 3x$. Vereenvoudig jou antwoord.

$$\begin{aligned} \frac{d}{dx} \sec 3x &= \frac{d}{dx} \frac{1}{\cos 3x} \\ &= \frac{-(-\sin 3x \cdot 3)}{\cos^2 3x} \\ &= 3 \frac{\sin 3x}{\cos 3x} \cdot \frac{1}{\cos 3x} \\ &= 3 \tan 3x \cdot \sec 3x \end{aligned}$$

2

Question 4 / Vraag 4

Determine the following limits, if they exist. / Bepaal die volgende limiete, indien hulle bestaan.

$$4.1 \ \lim_{x \rightarrow 1} \left(\frac{1}{\ln x} - \frac{x}{\ln x} \right) \quad (\infty - \infty)$$

$$= \lim_{x \rightarrow 1} \left(\frac{1-x}{\ln x} \right) \quad \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow 1} \left(\frac{-1}{1/x} \right)$$

$$= -1$$

2

$$4.2 \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x$$

$$(1^\infty)$$

$$\text{let } y = \left(1 + \frac{1}{x}\right)^x$$

$$\Rightarrow \ln y = x \ln \left(1 + \frac{1}{x}\right)$$

$$\Rightarrow \lim_{x \rightarrow \infty} \ln y = \lim_{x \rightarrow \infty} x \ln \left(1 + \frac{1}{x}\right) \quad (\infty \times 0)$$

$$= \lim_{x \rightarrow \infty} \frac{\ln \left(1 + \frac{1}{x}\right)}{\frac{1}{x}} \quad \left(\frac{0}{0}\right)$$

$$= \lim_{x \rightarrow \infty} \frac{-\frac{1}{x^2}}{1 + \frac{1}{x}} \div -\frac{1}{x^2} = 1$$

$$\Rightarrow \lim_{x \rightarrow \infty} y = e^1$$

3

$$4.3 \lim_{x \rightarrow \infty} \frac{x+8}{\sqrt{2x^2+3}}$$

$$= \lim_{x \rightarrow \infty} \frac{x \left(1 + \frac{8}{x}\right)}{|x| \sqrt{2 + \frac{3}{x^2}}}$$

$$\left(\sqrt{x^2} = |x| = x \text{ as } x > 0 \right)$$

$$= \lim_{x \rightarrow \infty} \frac{x \left(1 + \frac{8}{x}\right)}{x \sqrt{2 + \frac{3}{x^2}}}$$

$$= \frac{1}{\sqrt{2}}$$

3

Question 5 / Vraag 5

5.1 Complete the following: / Voltooi die volgende:

f is a decreasing function on an interval I if $f(x_1) > f(x_2)$ whenever $x_1 < x_2$ in I .

f is 'n dalende funksie op 'n interval I indien _____ wanneer $x_1 < x_2$ op I .

1

5.2 Use the definition in 5.1 to prove the theorem: If $f'(x) < 0$ on the interval I , then f is decreasing on I .

Gebruik die definisie in 5.1 en bewys die stelling: Indien $f'(x) < 0$ op die interval I , dan is f dalend op I .

Let x_1 and x_2 any two numbers in the interval I , with $x_1 < x_2$.

We are given $f'(x) < 0$

$\Rightarrow f$ diff. on (x_1, x_2) and
 f continuous on $[x_1, x_2]$

By the MVT there exists a

$c \in (x_1, x_2)$ so that

$$f'(c) = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

$$\Rightarrow f(x_2) - f(x_1) = f'(c)(x_2 - x_1)$$

Now $f'(c) < 0$ (given) on $x_2 - x_1 > 0$

$$\Rightarrow f(x_2) - f(x_1) < 0$$

$$\Rightarrow f(x_2) < f(x_1)$$

$\Rightarrow f$ is a decreasing function on I
Since x_1 and x_2 any two numbers.

3

Question 6 / Vraag 6

6.1 Show that $f(x) = \frac{e^x}{1-e^x}$ is an increasing function on the interval $[1, 2]$.

Toon aan dat $f(x) = \frac{e^x}{1-e^x}$ 'n stygende funksie is op die interval $[1, 2]$.

$$\begin{aligned} f'(x) &= \frac{e^x(1-e^x) - e^x(-e^x)}{(1-e^x)^2} \\ &= \frac{e^x}{(1-e^x)^2} \end{aligned}$$

$$\left. \begin{array}{l} e^x > 0 \quad \forall x \in \mathbb{R} \\ (1-e^x)^2 > 0 \quad \forall x \in \mathbb{R} \setminus \{0\} \end{array} \right\} \Rightarrow f'(x) > 0 \text{ on the interval } [1, 2]$$

2

6.2 Find the absolute maximum value of f on $[1, 2]$.

Vind die absolute maximum waarde van f op $[1, 2]$.

Abs max: $f(2) = \frac{e^2}{1-e^2}$

1